Alessio Salvatore

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Work Experience

Laelaps AI Jun 2026–Present
Robotics Engineer

- Working on the orchestration of robot fleets: developing low-level robot skills and orchestrating them with AI agents.
- As one such skill, built from scratch a multi-agent reinforcement learning pipeline — both the patrol simulator and the MAPPO (CTDE) training framework — that trains robot fleets for autonomous patrol of large areas.
- Kept the learned policy deployment-ready: it consumes only each agent's local observations and maps them end-to-end to velocity commands, and one policy transfers across different maps.
- Benchmarked policies intensively in the custom 2D simulator, with BenchMARL for different training algorithms, built on TorchRL for efficient GPU compute.
- Deployed and evaluated the trained policies on Unitree Go2 quadrupeds, scoring my framework against state-of-the-art approaches on the same scenarios.
- Built the automated data-collection and evaluation infrastructure behind that: batched GPU simulation, a fixed evaluation scenario suite, bit-for-bit reproducible runs, automated sweeps on self-terminating GCP GPU VMs, and a ROS 2 / Isaac Sim bridge.

ETH Robotics Club Nov 2025–Aug 2026
Partnerships Team Lead

- Secured sponsorships for club initiatives and events

JOIINT Lab — Italian Institute of Technology, Bergamo Feb–Jun 2025
Research Intern — Industrial Robotics Repo

- CAD design of a linear actuator built from commercial off-the-shelf components.
- Implemented real-time control algorithms using Arduino.
- Developed a MATLAB simulation to validate system dynamics and behavior.

Education

ETH Zürich, Zürich, Switzerland Sep 2025–Present
M.Sc. in Robotics
Coursework: Probabilistic Artificial Intelligence, Advanced Machine Learning, Image Analysis and Computer Vision, Planning and Decision Making for Autonomous Robots, 3D Vision, Perception and Learning for Robotics, Deep Generative Models, Robot Learning

Politecnico di Milano, Milan, Italy Sep 2022–Jul 2025
B.Sc. in Automation Engineering
Final Grade: 110/110 *cum laude* (equivalent GPA ≈ 4.0/4.0; 29.57/30).

Illinois Institute of Technology, Chicago, IL, USA Aug–Dec 2024
Exchange — Global E3 Program

Publications

PRoMiSe: Persistent Multi-Robot Monitoring Sim-to-Real-to-Field Project Page Nov 2026
A. Salvatore, J. de los Rios, D. Rytz, M. Stamatopoulou — *CoRL 2026 Workshop Sim2Real2Field*, Austin, TX

Projects

Research Projects

Knowledge Distillation for Aerial Robots Control Video Report Feb–Jun 2026
Autonomous Systems Lab – Roland Siegwart Lab (selected among 150 applicants for a 30-student cohort)

- Model-based controllers for an overactuated tilt-rotor MAV (OMAV, 10 actuators) rely on disturbance observers to handle model mismatch and external forces, but observer bandwidth, measurement noise, and tuning limit how fast they adapt
- Replaced it with a single recurrent LSTM policy distilling parameter-matched expert controllers over 30,000 randomized plants, trained closed-loop through differentiable vehicle dynamics on $\sim$42 h of simulated flight
- In simulation the policy outperforms the nominal model-based controller in velocity tracking ($p < 10^{-10}$, 2,000 paired rollouts) and cuts divergence under $\pm$30% model mismatch from 41.5% to 24%; deployed zero-shot on the OMAV

Goal-Driven Adaptive-Resolution 3D Mapping with RL Report Feb–Jun 2026

Computer Vision and Geometry Group – Marc Pollefeys Lab

- A robot mapping a building at uniform voxel resolution spends most of its memory on regions irrelevant to the task it was sent to do; framed online per-region resolution allocation (split/keep/merge) as an RL problem under an explicit memory budget
- Re-implemented PEANUT's dense voxel map as a sparse hierarchical multi-resolution grid (5/10/20 cm; Habitat, HM3D) and trained a PPO policy acting through two learned score thresholds, sidestepping an intractable per-block action space
- The learned policy retains 79% of a uniformly fine map's point-goal navigation success with 87% less memory (20 unseen episodes), matching an all-coarse map's memory footprint at far higher success

Selected Projects

Autonomous Exploration on a Unitree A2 Quadruped Repo 06/2026

ETH Robotics Summer School (ETHZ-RobotX)

- Deployed a fully autonomous search-and-explore mission with our team on a Unitree A2 quadruped across three field sites, exploring unknown environments to find and localize artifacts
- With the team, integrated and field-tuned the exploration stack – TARE and FAR planners with DLIO LiDAR-inertial odometry on a Hesai LiDAR – for real-robot deployment in tight spaces
- Developed the artifact-mapping node that fuses YOLO detections with TF2 transforms into a deduplicated map-frame log of artifact classes and 3D positions – the mission's scored output

Vision-Language-Action Policies on an SO-101 Arm Repo 05/2026

Project of the Class Robot Learning

- Investigated whether small open vision-language-action models (SmolVLA) can be fine-tuned to follow natural-language pick-and-place commands on a low-cost SO-101 arm, across three tasks of increasing linguistic difficulty
- Built the LeRobot training pipeline – prompt relabeling for linguistic variety, brightness and celebrity face-swap augmentation, and DAgger-style corrective demonstrations – with automated H100 cloud training streaming checkpoints to the Hugging Face Hub
- Final policies grounded color, compositional (ordinal, relative, negated), and celebrity-portrait references on the robot at 10 FPS, including generalization to celebrity identities never seen in training

Person Following on the Agibot X2 Humanoid Repo 05/2026

RoboHack Hackathon

- Built real-time person detection and following on the Agibot X2 humanoid in a hackathon weekend, running fully onboard: YOLOv8 on the head stereo cameras with disparity-based depth giving the target's 3D position
- Closed the loop with Ruckig head tracking and a state-machine locomotion supervisor (approach / stop-band / align) holding a set distance, deployed on the robot behind dry-run and gantry test stages and a 0.5 s velocity watchdog

Channel Position Estimation from Pipe-Inspection Video Repo 05/2026

Physical AI Hackathon

- Estimated per-frame absolute position, turning-point frame, and movement direction of a sewer-inspection probe traveling out and back through a channel of known length, using its video feed alone – no other sensors
- Built a classical-CV pipeline – sparse KLT optical flow with MAD outlier filtering, triple scene gating before integration, and scale-free dual-leg normalization – processing $\sim$1000 frames in under 8 s, CPU-only

DINOv3 for Mitral Valve Segmentation Repo 12/2025

- Segmented the mitral valve in echocardiography videos where only three frames per training clip are annotated, propagating DINOv3 patch features to turn the sparse annotations into dense per-frame pseudo-labels
- Built a unified preprocessing pipeline for the heterogeneous expert- and amateur-annotated video data and trained a U-Net segmentation model on the filtered pseudo-labels

Multi-Agent Multi-Goal Collection Planner 12/2025

Project of the Class Planning and Decision Making for Autonomous Robots

- Built a planning framework for a team of robots collecting multiple goals in a partially observable environment, using Large Neighborhood Search to allocate goals across robots.
- Combined the allocation with real-time path planning and decentralized collision avoidance, so each robot replans and avoids the others as the environment is revealed.

Autonomous Satellite Docking Planner 11/2025 – 12/2025

Project of the Class Planning and Decision Making for Autonomous Robots

- Built a trajectory optimization pipeline for autonomous satellite docking in cluttered environments, where obstacle avoidance makes the guidance problem non-convex.
- Solved it via Sequential Convexification, iteratively optimizing convexified subproblems to produce trajectories that steer the spacecraft past obstacles to its docking target in the course simulator.